

Ravi Prakash Srivastava

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Education

Indian Institute of Information Technology Ranchi (IIIT Ranchi), India

August 2024 – April 2026

M.Tech (Data Science & AI)

CGPA: 8.45 (ongoing)

Institute of National Importance, Govt. of India

North Eastern Hill University (NEHU), Shillong, India

August 2018 – June 2022

B.Tech (Information Technology)

CGPA: 7.85

Central University, Govt. of India

Publications

- **Adversarial Prompt Injection Attacks on LLM: Cryptographic Key Leakage and Defense Strategies**
Under Publication at International Conference on Advanced Computing and Applications (ICACA-2026).
- **IoT-HITS: IoT-based Human Intrusion Detection System for Border Regions using Deep Learning**
[IETE Journal of Research] DOI: <https://doi.org/10.1080/03772063.2025.2521688>
- **Deepfake Video Detection Using Face-Centric Processing and Frame Sampling**
Under Publication International Conference on Advanced Computing and Applications (ICACA-2026).
- **Balanced Few-Shot Episodic Learning for Accurate Retinal Disease Diagnosis**
(arXiv: <https://arxiv.org/abs/2512.04967v1>)
- **Smart City Vehicle Accident Monitoring and Detection System Using MEMS, GSM, GPS Raspberry Pi 4**
[IETE Journal of Research] DOI: <https://doi.org/10.1080/03772063.2022.2043787>

Experience

Software Engineer SS Software Solutions LLC , Hyderabad

March 2026 – Present

- Collaborated with stakeholders to translate business requirements into technical AI/ML solutions, driving design decisions from prototype to deployment.
- Defined and enforced ML pipeline standards including versioning, logging, and monitoring to improve system reliability and debuggability.

Research Data Scientist Pantech e-learning, India

February 2023 – September 2024

- Designed and deployed predictive machine learning models with emphasis on robustness and reliability under noisy and incomplete real-world data.
- Conducted systematic robustness evaluation using stress testing and distributional shift analysis to identify failure modes and inform mitigation strategies.
- Built end-to-end data preprocessing pipelines to handle missing values, class imbalance, and sensor noise, reducing downstream model error by improving input data quality across multiple project domains.

Machine Learning Intern (C-DAC), India

August 2022 – February 2023

- Performed large-scale data preprocessing and feature engineering on student activity logs (attendance, assignments, quizzes) to support predictive modeling.
- Implemented and benchmarked multiple machine learning models, including Decision Trees, Random Forests, and Support Vector Machines, for academic performance prediction.

Projects

Master's Thesis: Adversarial Prompt Injection Attacks on LLMs: Cryptographic Key Leakage and Defense Strategies:

- Designed and evaluated attack scenarios where injected prompts manipulate LLM behavior to expose sensitive information embedded in system prompts or context windows.

- Proposed and benchmarked defense strategies including prompt sanitization, input/output filtering, and instruction hierarchy enforcement to mitigate injection-based leakage risks.

Bachelor's Thesis: IoT-HITS: IoT-based Human Intrusion Detection System for Border Regions Using Deep Learning:

- Developed an IoT-based border security system using deep learning for real-time human intrusion detection.
- Designed the system architecture, integrated sensors, and implemented the deep learning detection model with alert mechanisms.

Technical Skills

- **Machine Learning & Security:** Deep Learning, Transformer Models, Adversarial Machine Learning, Prompt Injection Attacks, Deepfake Detection, Representation Learning, HPC ,
- **Tools & Frameworks:** PyTorch, Hugging Face Transformers, Scikit-learn, NumPy, Pandas.
- **Programming:** Python, C++, SQL, Bash.
- **Security & Research Tools:** Linux, Git, Jupyter Notebook, LaTeX.

Achievement

- Received the Institution of Electronics and Telecommunication Engineers (IETE) Scholarship for academic excellence during M.Tech studies in Computer Science and Engineering.
- Best Paper Award, ICACA 2026 for the paper *"Adversarial Prompt Injection Attacks on Large Language Models: Cryptographic Key Leakage and Defense Strategies"*.

Professional Affiliations

- The Institution of Electronics and Telecommunication Engineers (IETE) – Associate Member [AM-503427]